

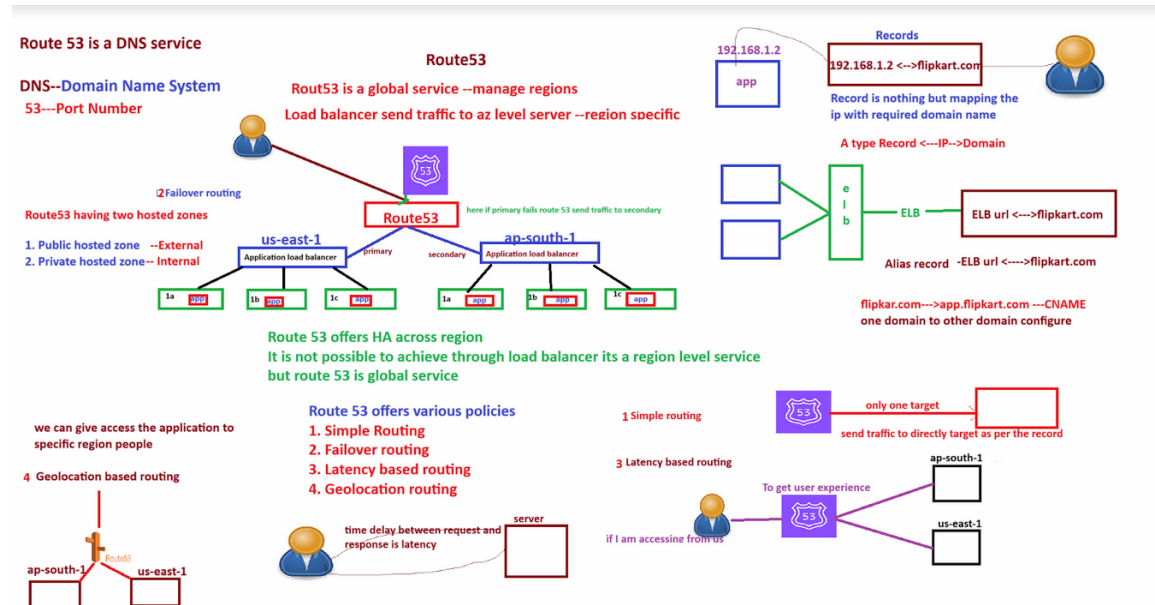
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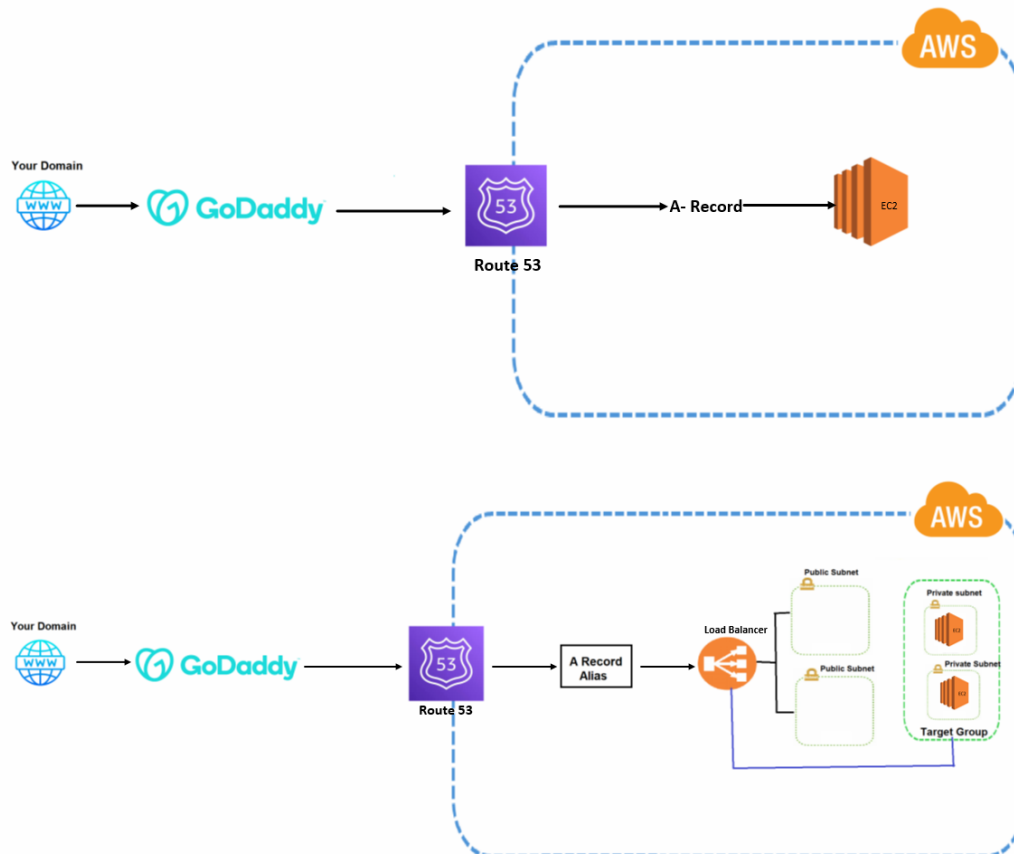
VEERA

Steps to Configure Route 53

Diagram - Class



Diagram



High-level steps

1. Create a VPC.
2. Create an Internet Gateway and attach it to the VPC.
3. Create one public subnets: one in Availability Zone 2a
4. Create a route table
5. Associate the Internet Gateway, and link it to the public subnets in Availability Zones 2a
6. Create a security group with inbound rules for SSH and HTTP access.
7. Launch an EC2 instance in the public subnet in AZ 2a
8. Connect to public server and install httpd

```
sudo su
sudo yum install httpd -y
sudo systemctl start httpd
sudo systemctl enable httpd
cd /var/www/html
vi index.html
<h1>Welcome </h1>
ESC then :wq
Hit enter
```

9. Create your domain name
10. Create hosted zone in route 53
11. Create A type Record in hosted zone

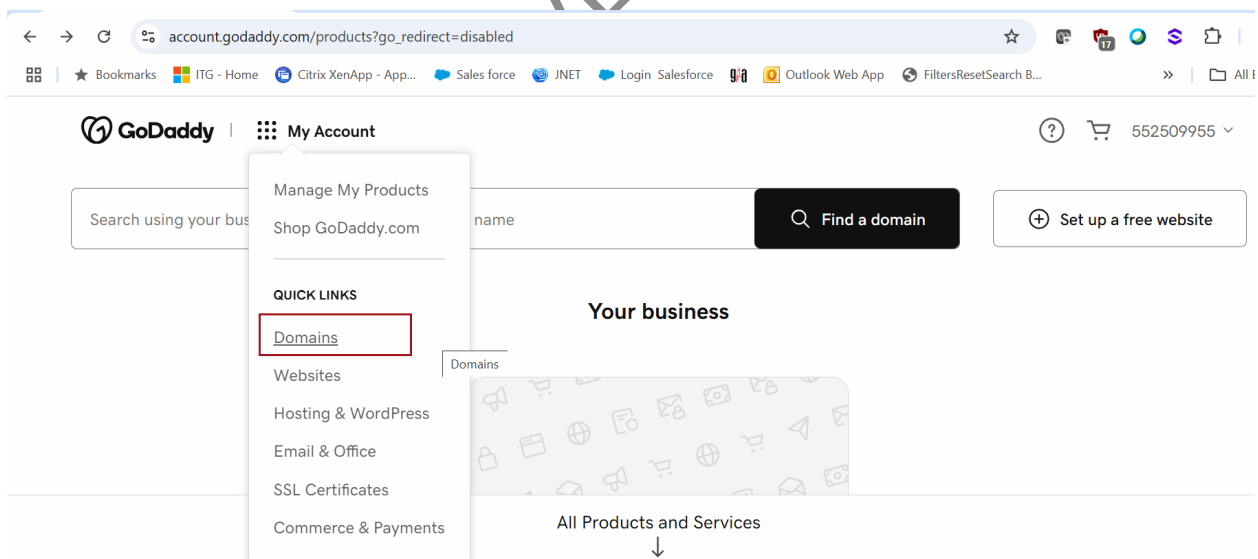
VEERA

Screenshots to see Name servers of your Domian

step-by-step guide (with example screenshots) to create an **A record** in Route 53 to route traffic directly to an **EC2 instance** using its public IP address. This setup assumes that your application is accessible via the EC2 instance's public IP

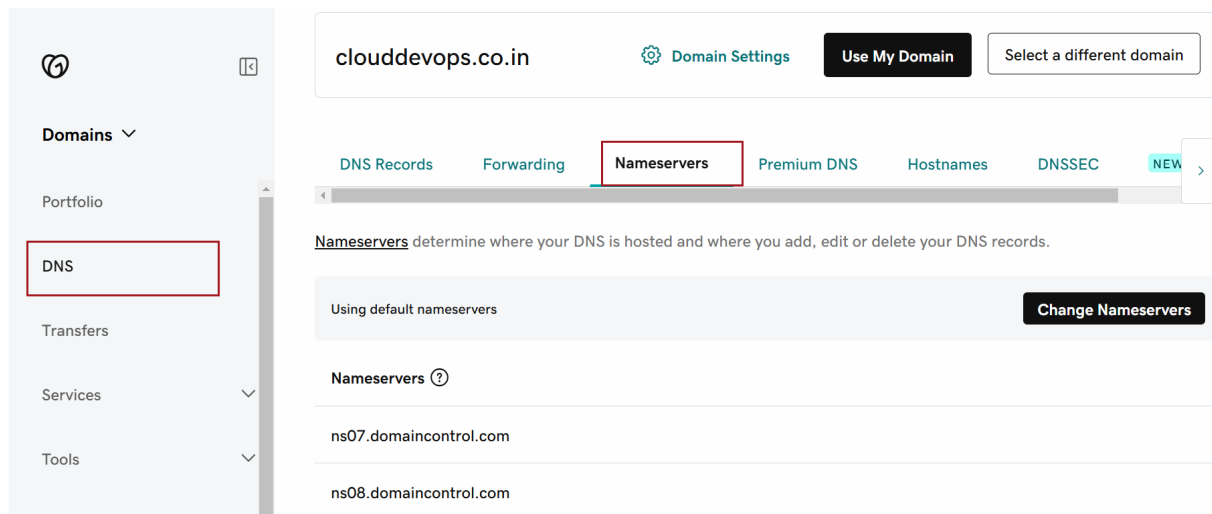
After signing in, click on **My Products** in the top-right corner.

Scroll down to **Domains**, and find the domain for which you want to view the name servers



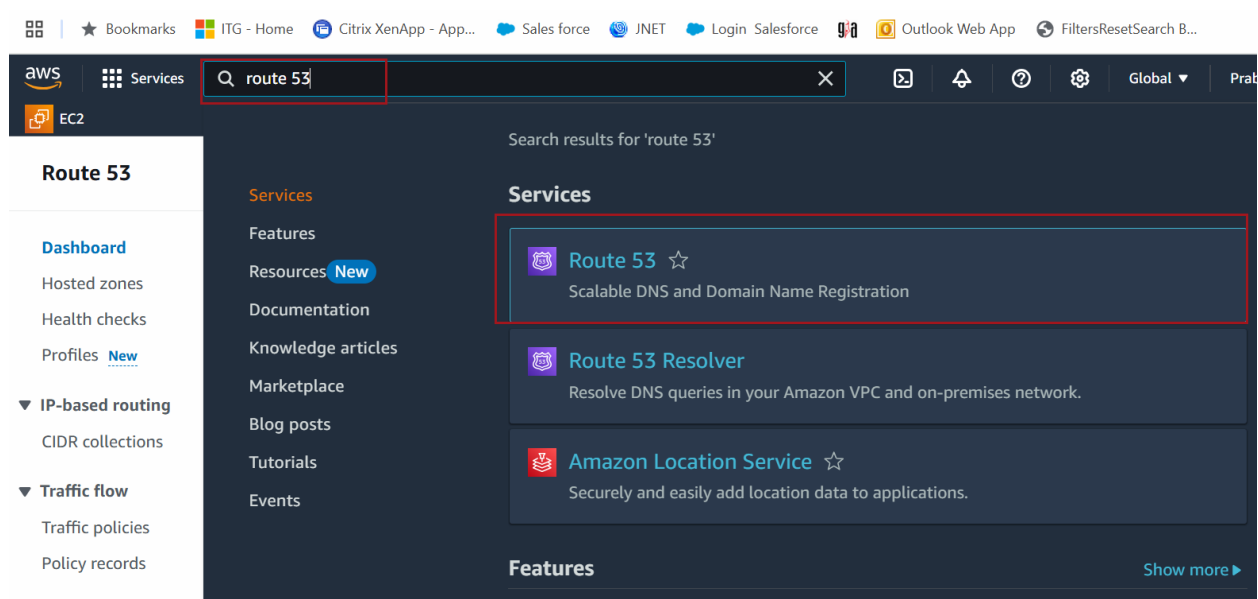
Click on **DNS** to open the DNS management page

Under the **Nameservers** section, you will see the current name servers for your domain.



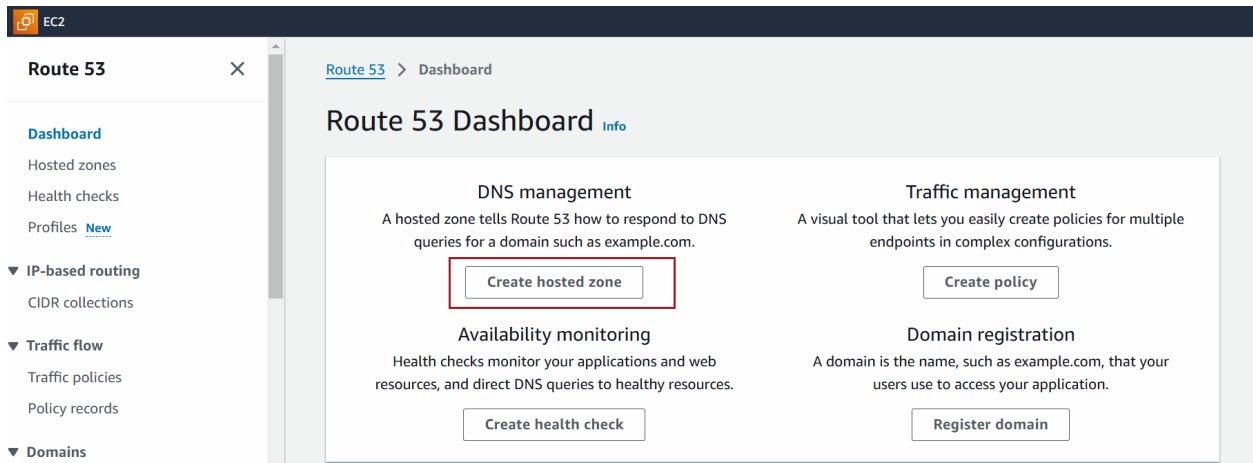
Screenshots for hosted zone creation

Go to the Route 53 Console



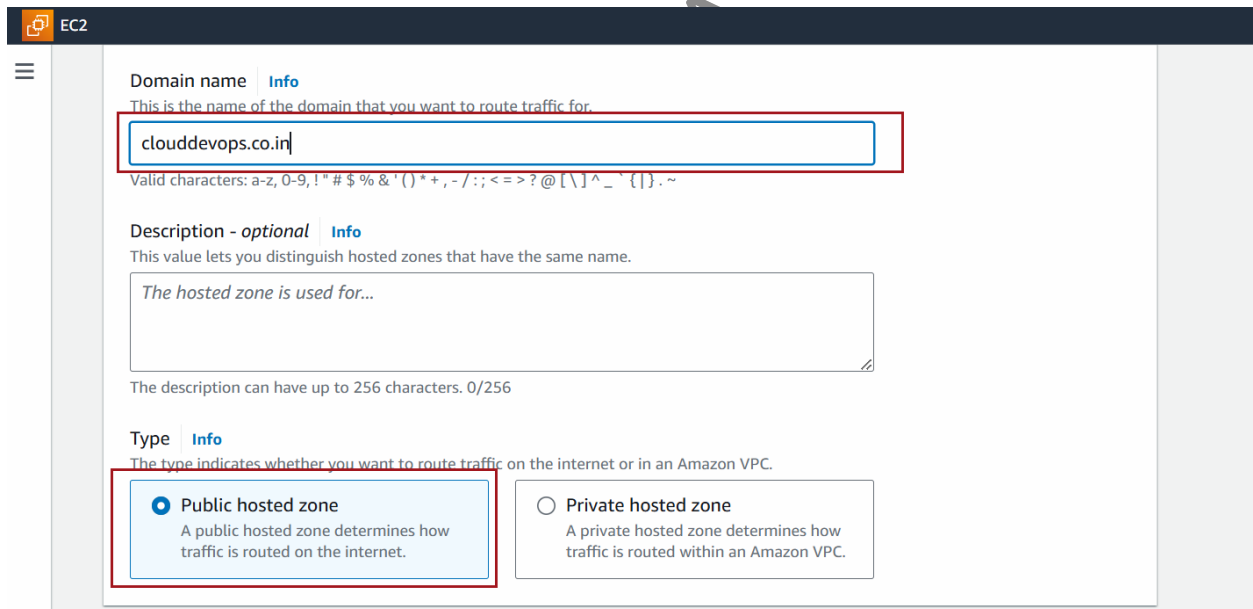
In the **DNS management** section, choose **Hosted Zones**.

Click **Create hosted zone**



Enter the **Domain name**

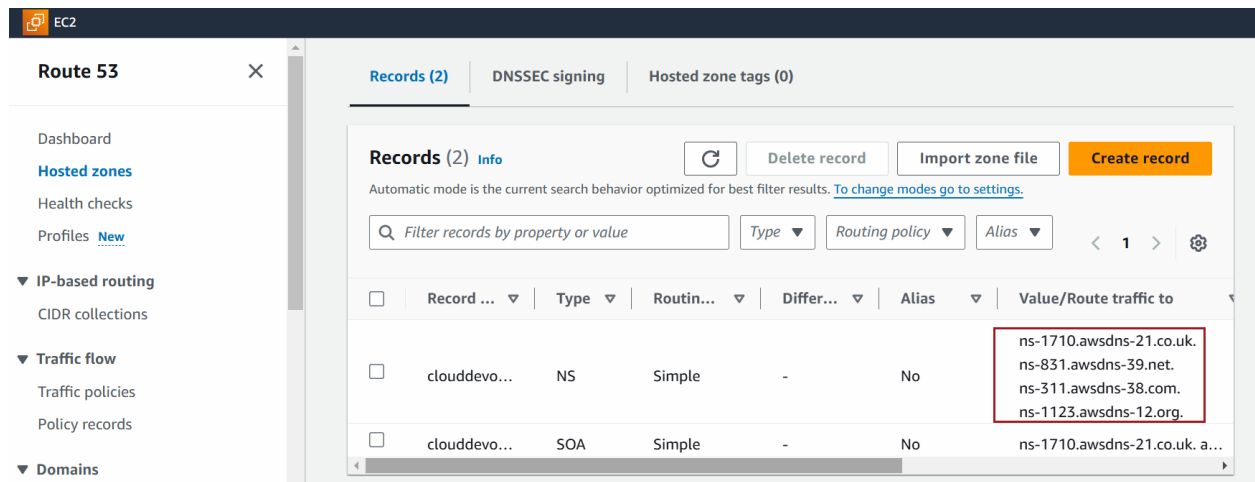
Specify the type as **Public Hosted Zone**



Update Domain Name Servers (DNS) with Registrar

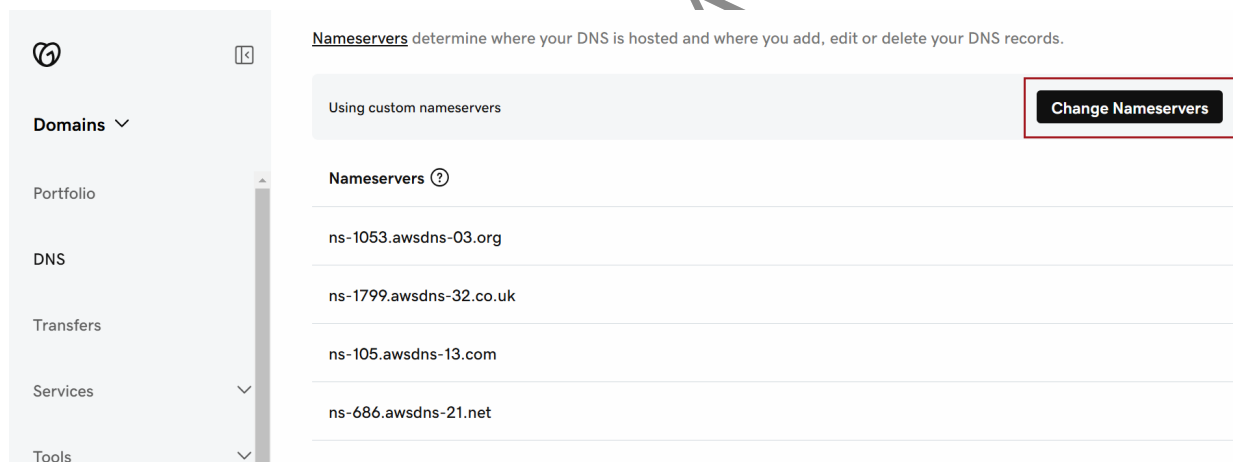
If you purchased your domain from another registrar (e.g., GoDaddy), update the name servers to the ones provided by Route 53.

Copy the name servers from route53 and update on your domain. When copying each name server, simply exclude the final period (".") if it appears at the end.



Screenshots to update the name servers on your domain

Choose Change Nameservers on your domain



Enter nameservers from route53

Click on Save

Edit nameservers
Choose nameservers for clouddevops.co.in

☐ GoDaddy Nameservers (recommended)

☒ I'll use my own nameservers

ns-1710.awsdns-21.co.uk

ns-831.awsdns-39.net

ns-311.awsdns-38.com

ns-1123.awsdns-12.org

[+ Add Nameserver](#)

Save **Cancel**

Screenshots for Simple Record Policy

Configure DNS Records

After creating the hosted zone, add DNS records to route traffic to EC2 instance

Click **Create Record** in the hosted zone

Route 53

Dashboard
Hosted zones
Health checks
Profiles [New](#)

▼ IP-based routing
CIDR collections

▼ Traffic flow
Traffic policies
Policy records

▼ Domains

Records (2) DNSSEC signing Hosted zone tags (0)

Records (2) [Info](#) [Refresh](#) [Delete record](#) [Import zone file](#) **Create record**

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

[Type](#) [Routing policy](#) [Alias](#) [1](#) [Settings](#)

☐	Record ...	Type	Routin...	Differ...	Alias	Value/Route traffic to
☐	clouddevo...	NS	Simple	-	No	ns-1710.awsdns-21.co.uk. ns-831.awsdns-39.net. ns-311.awsdns-38.com. ns-1123.awsdns-12.org.
☐	clouddevo...	SOA	Simple	-	No	ns-1710.awsdns-21.co.uk. a...

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In the **Record type** dropdown, select **A**

In the **Value** field, enter the public IP address

EC2

Create record [Info](#)

[Switch to wizard](#)

▼ Record 1 [Delete](#)

Record name [Info](#) clouddevops.co.in

Keep blank to create a record for the root domain.

☐ Alias

Value [Info](#)

Record type [Info](#)

Enter multiple values on separate lines.

Choose **Routing policy** as Simple routing.

Click **Create records**

EC2

Create record [Info](#)

[Add another record](#)

Value [Info](#)

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

Recommended values: 60 to 172800 (two days)

Routing policy [Info](#)

Test Accessing Your Application

Wait a few minutes for DNS changes to propagate

Open **Internet Explorer** or any web browser.

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In the address bar, type your domain name and press **Enter**

If Route 53 and your DNS configuration are correct, your website or service should load as expected.

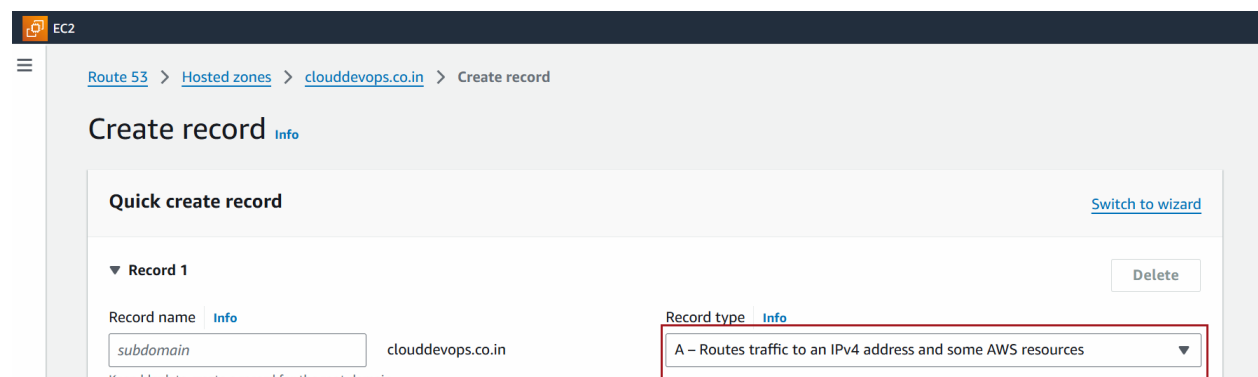


Screenshots for Sime Record Creation with Alias option

Step-by-step guide with screenshots to create an Alias record in Route 53 to route traffic to your load balancer once you've confirmed that your application is accessible using the load balancer's DNS name.

In the hosted zone for your domain, click on **Create Record**

Record type: Choose **A - IPv4 address**



Alias: Select **Yes** to enable the alias

Route traffic to: Choose **Alias to Application and Classic Load Balancer.**

Region: Select the AWS region where your load balancer is deployed

Choose Load Balancer: In the dropdown, select your load balancer from the list

Click **Create records**

EC2

Keep blank to create a record for the root domain.

☒ Alias

Route traffic to [Info](#)

Alias to Application and Classic Load Balancer

US West (Oregon)

Alias hosted zone ID: Z1H1FLSHABSF5

Routing policy [Info](#)

Simple routing

Evaluate target health ☒ Yes

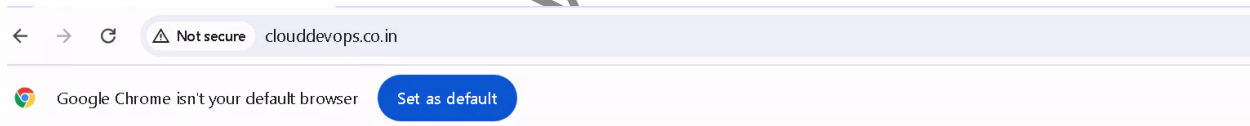
[Add another record](#)

[Cancel](#) [Create records](#)

Test Accessing Your Application

Wait a few minutes for DNS changes to propagate

Open a browser and go to your custom domain. It should now route to your application via the load balancer.



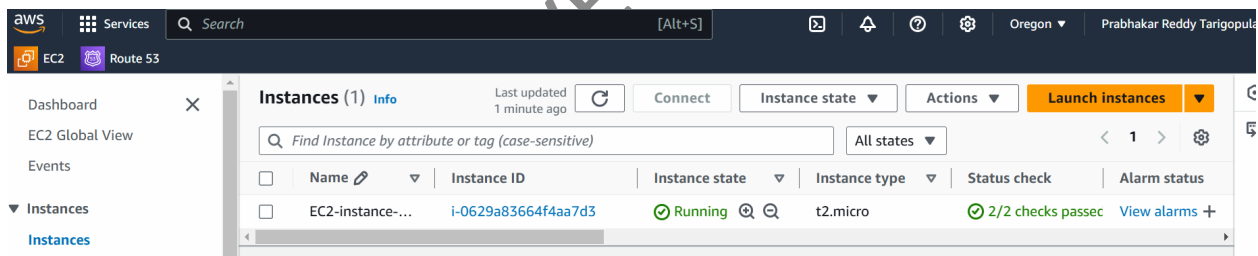
Welcome to my EC2 instance

Screenshots for Failover Record Policy

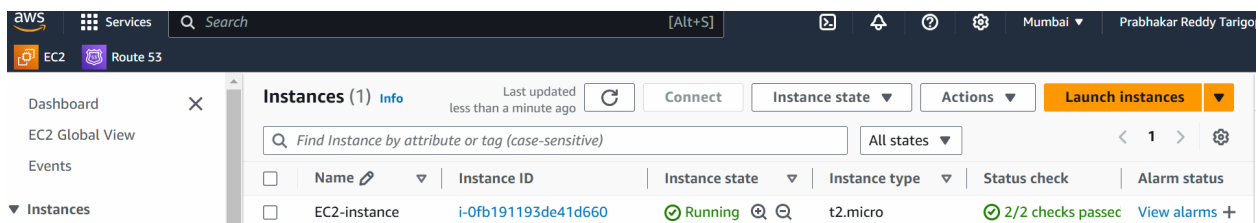
Define Primary and Secondary Resources

Identify the primary resource (server,) and secondary resource (backup IP or server)

Created one instance under Oregon Region



Created another instance under Mumbai Region

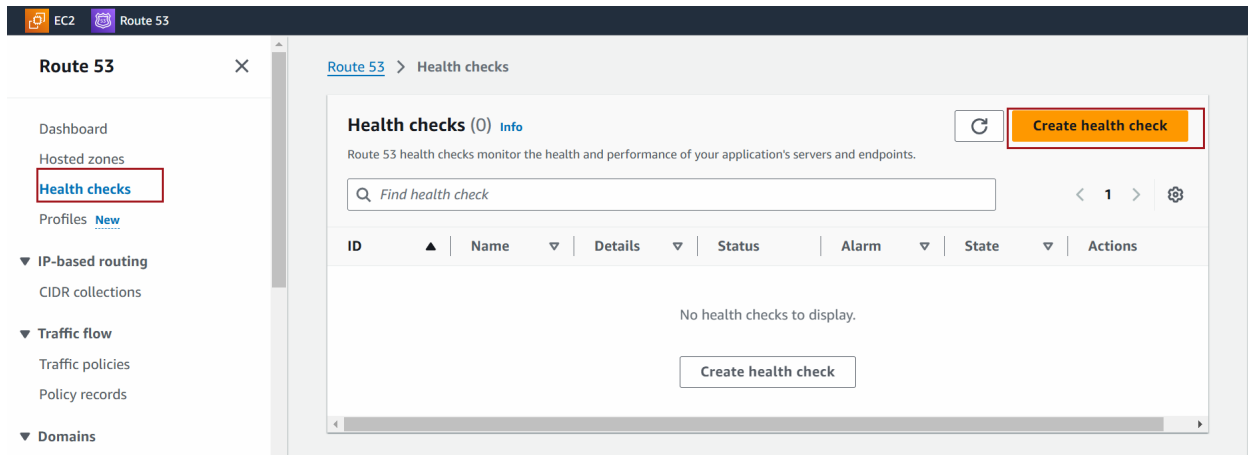


Set Up Health Checks

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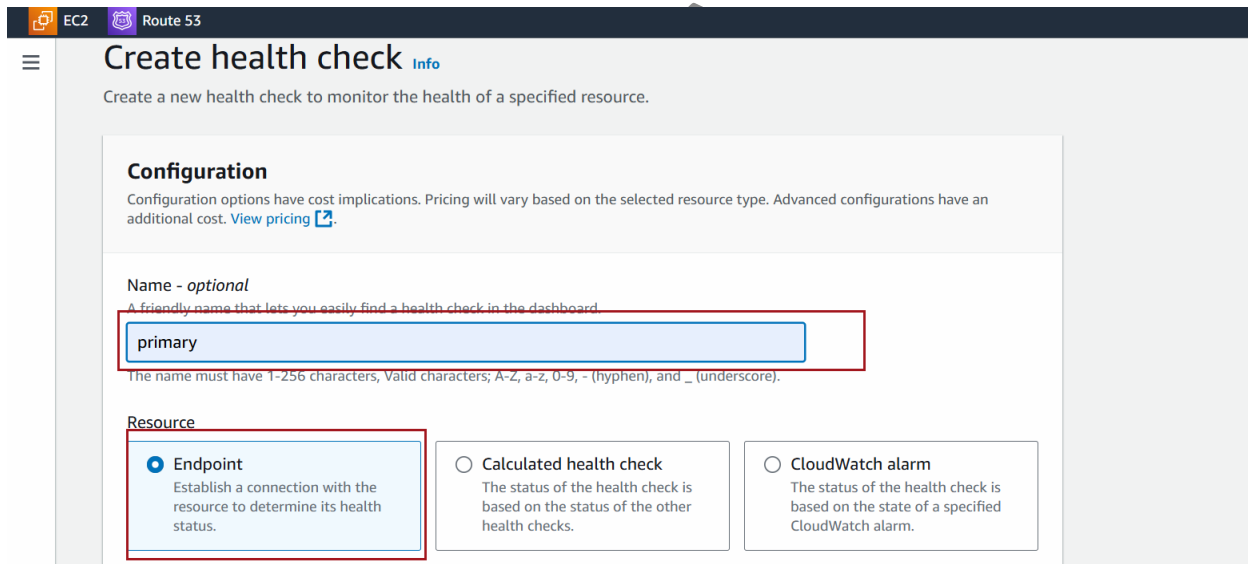
From the left navigation pane, click on **Health Checks**.

Click **Create health check**.



Name: Enter a unique name for the health check.

Select Resource - Endpoint



Specify Endpoint: IP Address

IP address: Provide the primary resource's IP address

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Specify endpoint by

☒ IP address

☐ Domain name

IP address

The path can be any value for which your endpoint will return an HTTP status code of 2xx or 3xx when the endpoint is healthy.

HTTP ▼ 13.201.98.155

► Advanced configuration

click **Create health check**.

IP address

The path can be any value for which your endpoint will return an HTTP status code of 2xx or 3xx when the endpoint is healthy.

HTTP ▼ 13.201.98.155

► Advanced configuration

Tags - optional

A tag is a label that you assign to an AWS resource. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tags

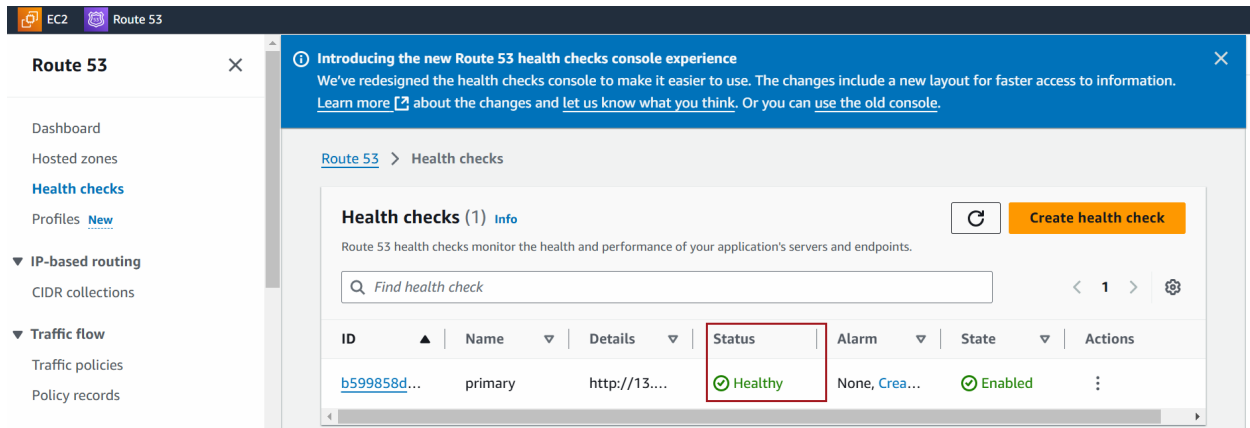
You can add up to 50 more tags.

Cancel Create health check

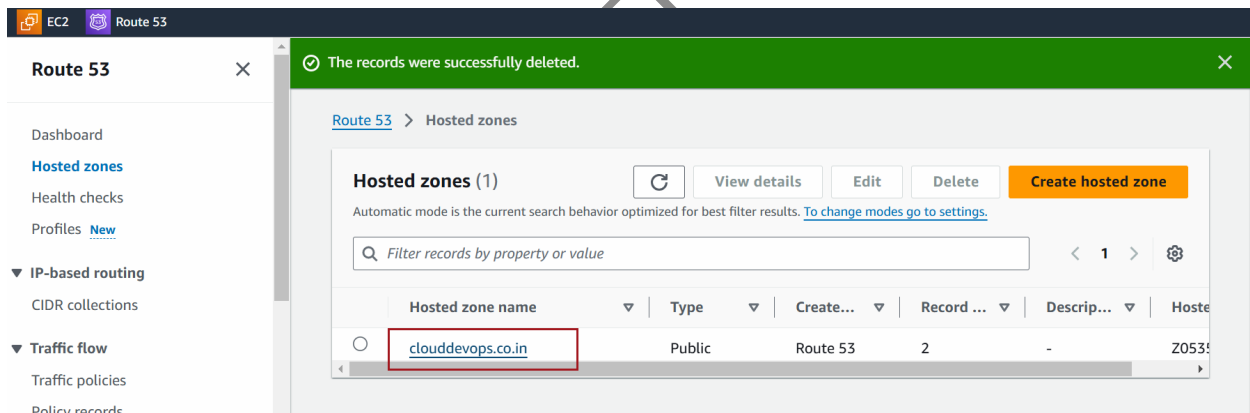
Wait for a few minutes; it typically takes around 2–3 minutes for Route 53 to complete the initialization phase.

After this, the health check will either show **"Healthy"**

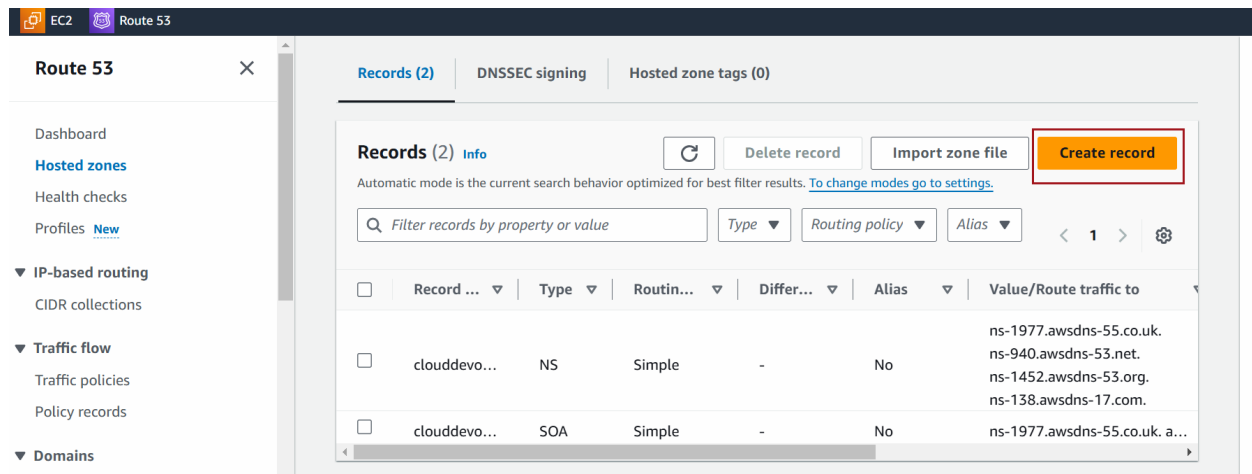
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From the Route 53 console, go to **Hosted Zones** and select your domain



lick **Create record**.



Configure Primary Record:

Record Type: Choose A

Value: Enter the IP address

Create record [Info](#)

Quick create record [Switch to wizard](#)

▼ **Record 1** [Delete](#)

Record name [Info](#)
subdomain cloudevops.co.in
Keep blank to create a record for the root domain.

☐ **Alias**

Record type [Info](#)
A - Routes traffic to an IPv4 address and some AWS resources ▼

Value [Info](#)
54.188.72.174

Enter multiple values on separate lines.

Routing Policy: Choose Failover.

Failover Record Type: Set this as **Primary**

Associate Health Check: Select the health check you created for the primary resource

Record ID – Enter record ID

Click **Create records** to save

TTL (seconds) [Info](#)
300 1m 1h 1d
Recommended values: 60 to 172800 (two days)

Routing policy [Info](#)
Failover ▼

Failover record type
Choose **Primary** to route traffic to the specified resource by default or **Secondary** to route traffic to the specified resource when the primary resource is unavailable. You can create only one failover record of each type.
Primary ▼

Health check ID [Info](#)
Q b599858d-4ef0-4503-8b4a-8e8c7e0aa147 X ↺

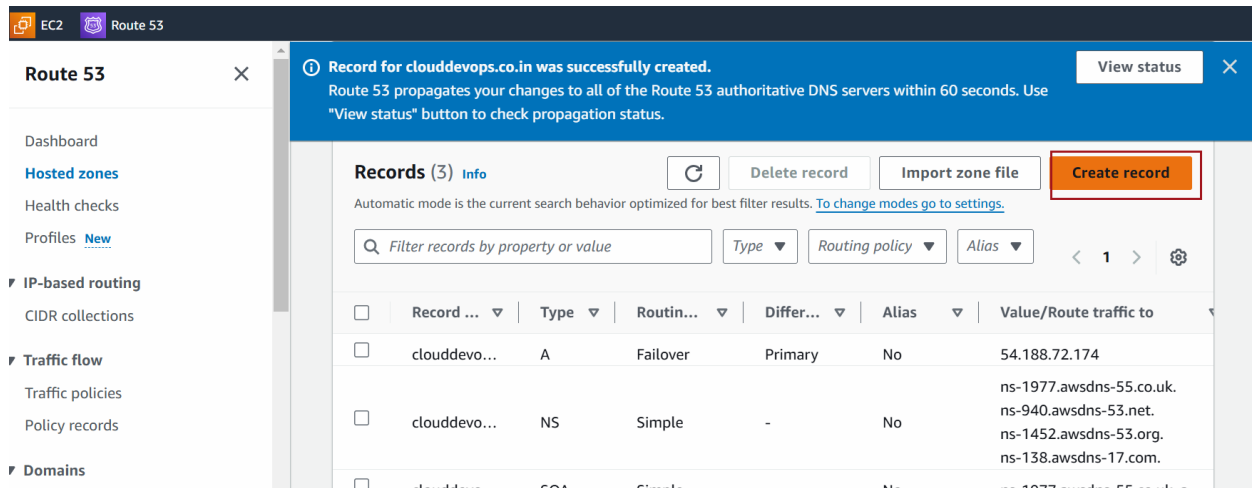
Record ID [Info](#)
RID-1

[Add another record](#)

[Cancel](#) [Create records](#)

Create the Secondary Failover Record

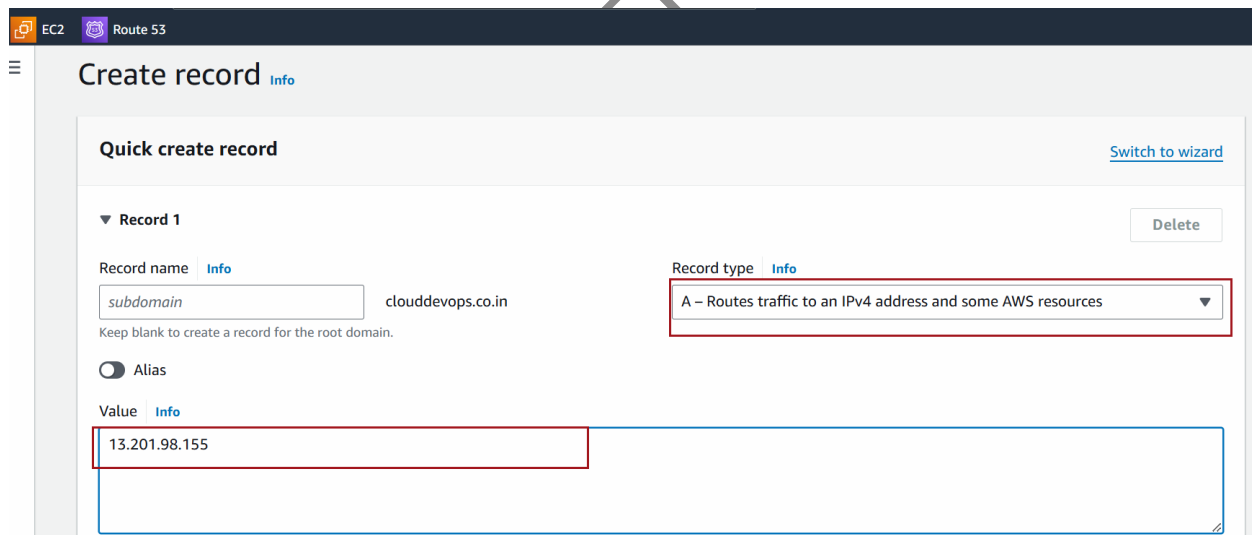
Click **Create record again** to add the secondary failover record



Configure Secondary Record:

Record Type: Use the same record type as the primary

Value: Enter the IP address or DNS name of the secondary



Routing Policy: Choose **Failover**.

Failover Record Type: Set this as **Secondary**

Record ID: Any ID

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

300 1m 1h 1d

Recommended values: 60 to 172800 (two days)

Routing policy [Info](#)

Failover

Failover record type

Choose Primary to route traffic to the specified resource by default or Secondary to route traffic to the specified resource when the primary resource is unavailable. You can create only one failover record of each type.

Secondary

Health check ID - optional [Info](#)

Choose health check

Record ID [Info](#)

RID-2

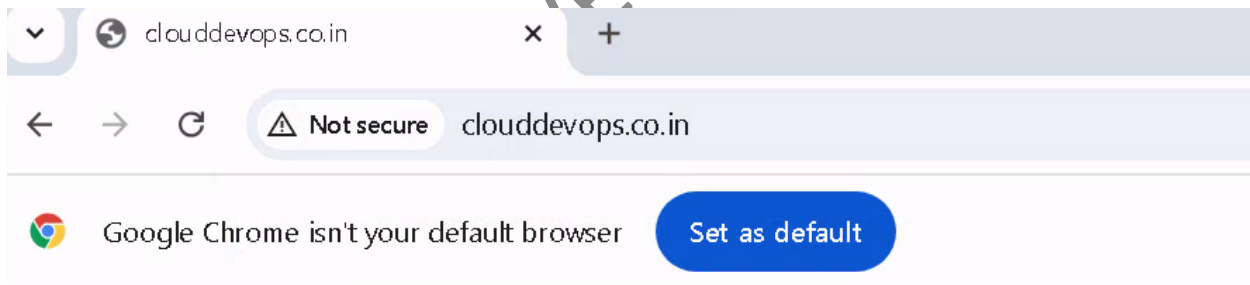
Add another record

Cancel Create records

Test the Failover Configuration

Wait a few minutes for DNS changes to propagate

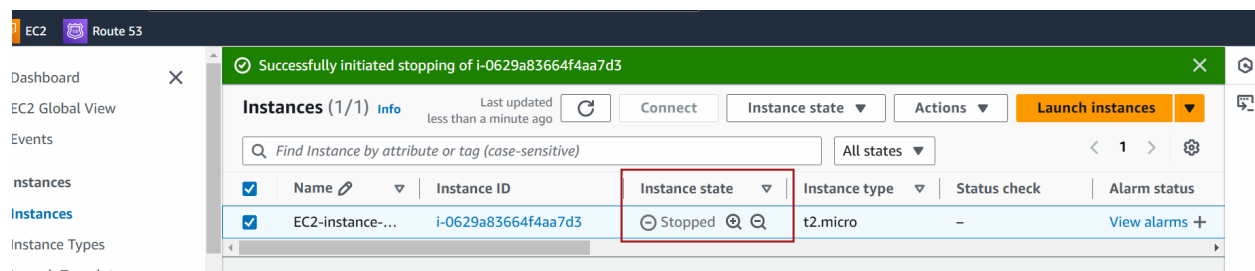
Open a browser and go to your custom domain. It should now route to your application via the load balancer.



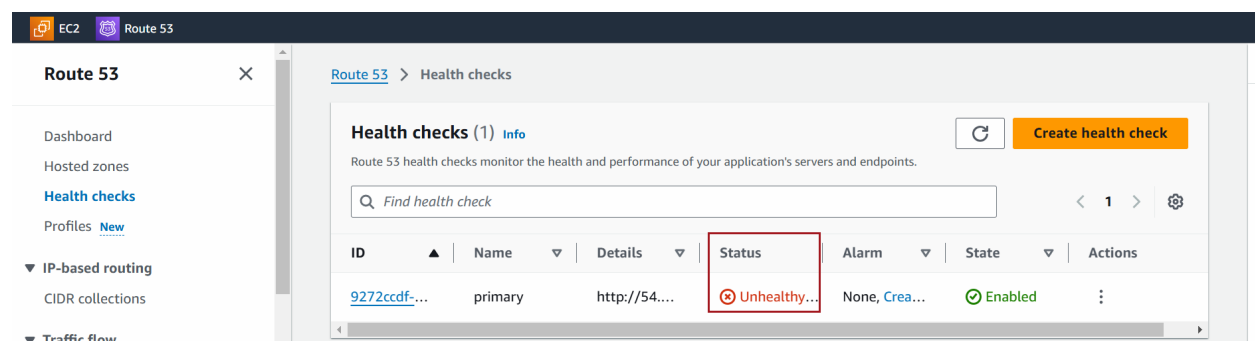
EC2 instance from US(Oregon)

Temporarily take the primary resource offline

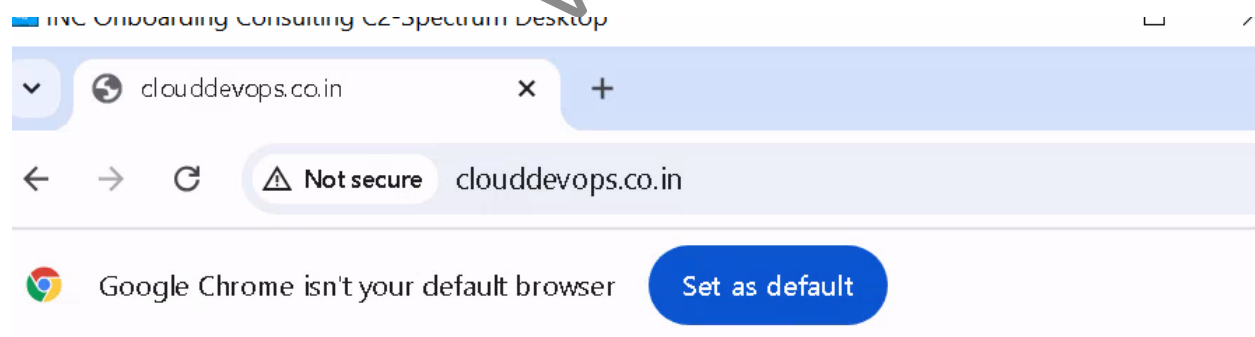
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Check if Route 53 detects the primary as unhealthy



Check if Route 53 detects the primary as unhealthy and automatically routes traffic to the secondary.



EC2- Instance from Mumbai

Screenshots for Geolocation Record Policy

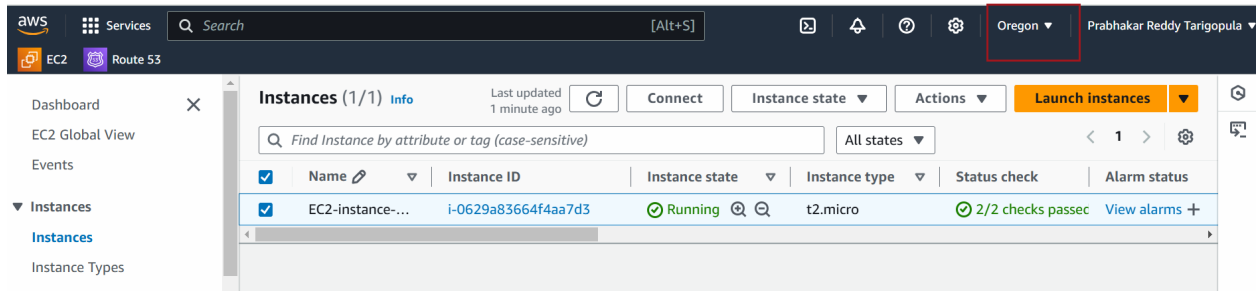
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End Resources Set Up to Handle Location-Based Traffic

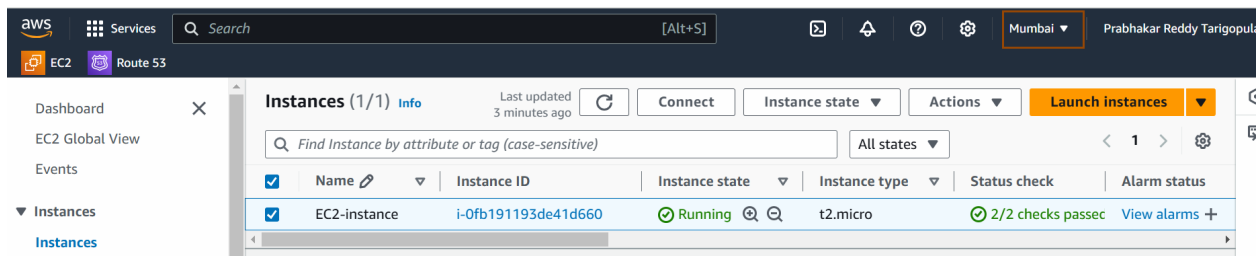
Ensure you have different servers, for each geographic location where you want to direct traffic.

For example, if you want to route North American users to a US-based server and European users to an EU-based server, set up these resources in advance

Created an instance in US WEST(Oregon)

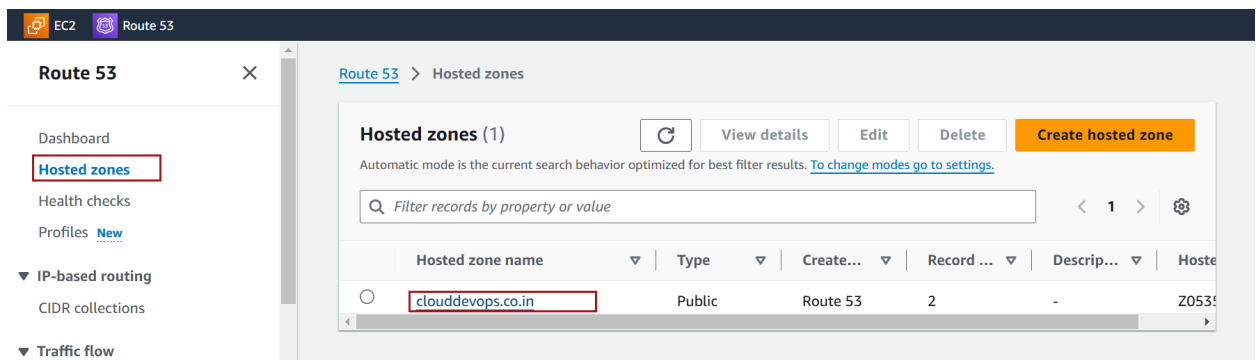


Created another instance in Asia Pacific (Mumbai)



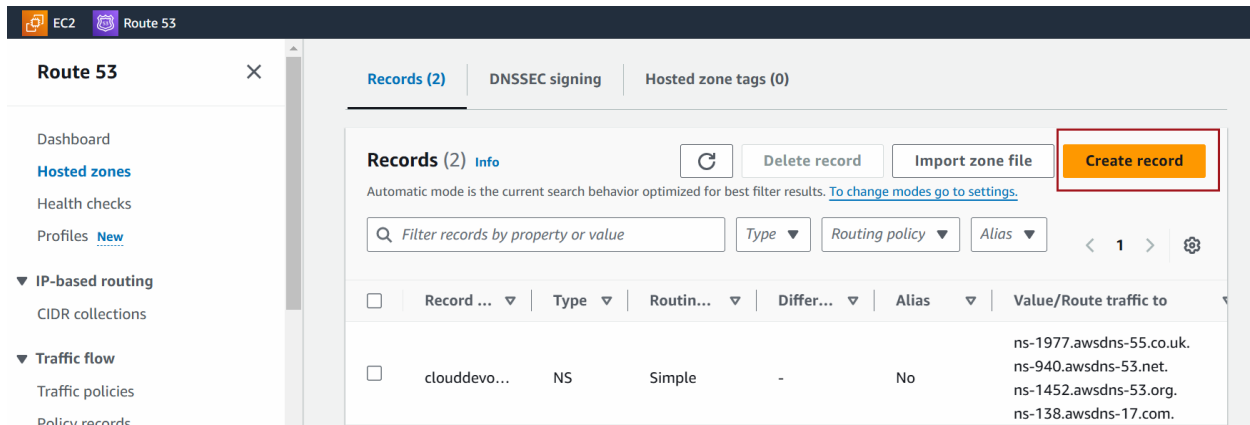
In the Route 53 dashboard, go to **Hosted Zones**.

Select the hosted zone for the domain you want to configure geolocation-based routing for.



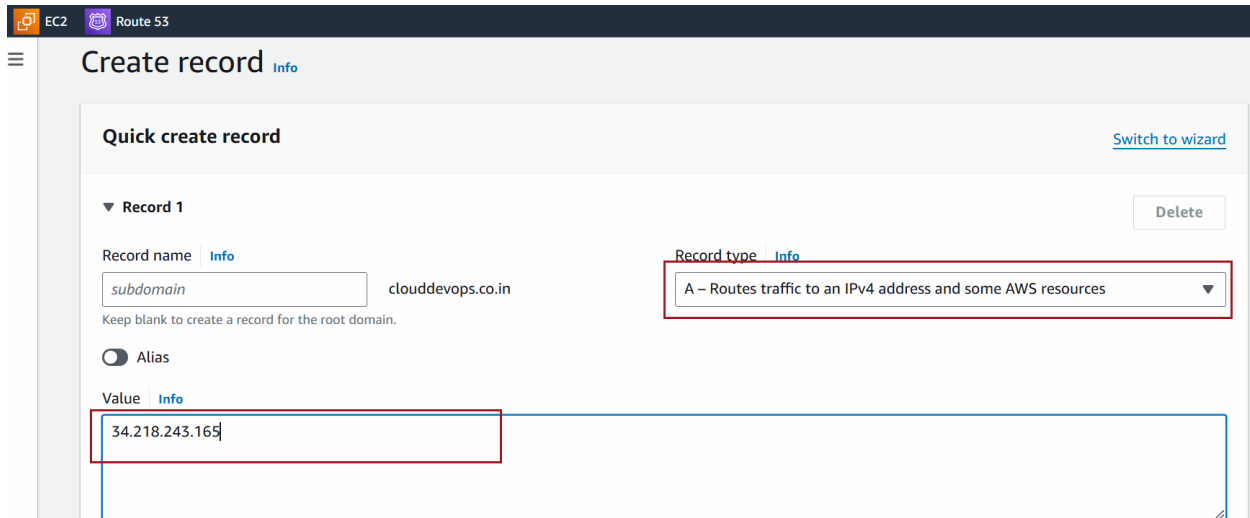
/

Click **Create record**.



Record Type: Choose the record type as **A**

Value: Enter the public IP address of the instance created in US WEST(Oregon)



Routing Policy: Select **Geolocation**.

Location:

- Choose the location to route traffic based on user geography:
 - **Continent:** E.g., North America, Europe.
 - **Country:** E.g., United States, India.
 - **State:** For the United States, you can also choose a specific state (e.g., California).

Click **Create records** to save

TTL (seconds) [Info](#)

300 1m 1h 1d

Recommended values: 60 to 172800 (two days)

Routing policy [Info](#)

Geolocation

Location

Choose a location that DNS queries originate from. Queries from that location are routed to the resource that you specify in this record. Choose **Default** to route queries from locations that you don't create records for.

North America

Record ID [Info](#)

RID-1

Health check ID - optional [Info](#)

Choose health check

Add another record

Cancel Create records

Add Additional Geolocation Records (Optional)

Repeat the above steps for other geographic location Asia Pacific(Mumbai) you want to route traffic to. For example, you might create records for different continents or countries.

Route 53

Dashboard

Hosted zones

Health checks

Profiles [New](#)

IP-based routing

CIDR collections

Traffic flow

Traffic policies

Policy records

Domains

Records (4) [Info](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

Type Routing policy Alias

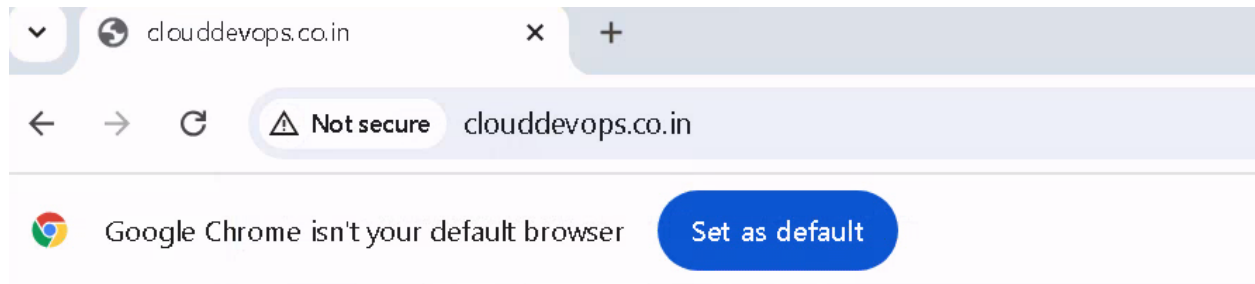
	Record ...	Type	Routin...	Differentiator	Alias	Value/Route traffic to
☐	clouddevo...	A	Geolocation	North America	No	34.218.243.165
☐	clouddevo...	A	Geolocation	Asia	No	3.108.54.77
☐	clouddevo...	NS	Simple	-	No	ns-1977.awsdns-55.co. ns-940.awsdns-53.net. ns-1452.awsdns-53.org ns-138.awsdns-17.com
☐	clouddevo...	SOA	Simple	-	No	ns-1977.awsdns-55.co.

Wait a few minutes for DNS changes to propagate

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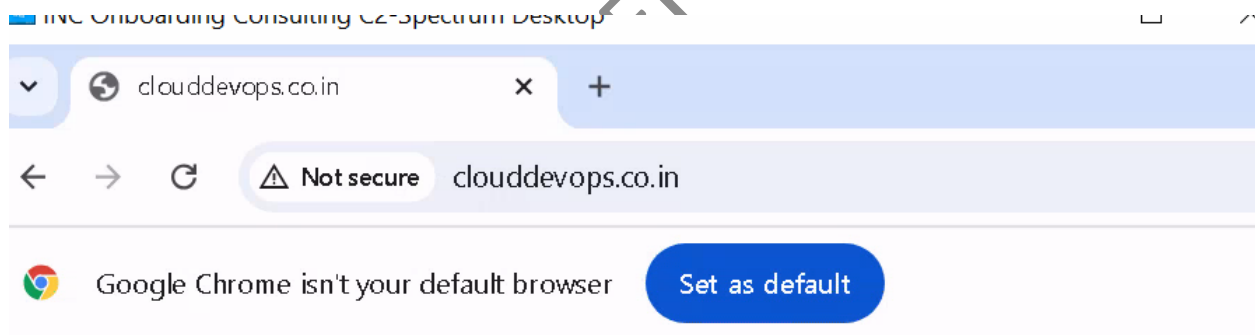
Open a browser and go to your custom domain on the server from the **same continent**. It should now route to your application via the load balancer.

From North America location server



EC2 instance from US(Oregon)

From Asia Pacific location server



EC2- Instance from Mumbai

Screenshots for Latency-Based Record policy

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For each AWS region you want to route traffic to based on latency, follow these steps:

Record Type: Choose **A**

Value: Enter the public IP address

EC2 Route 53

Create record [Info](#)

Quick create record [Switch to wizard](#)

▼ Record 1 [Delete](#)

Record name [Info](#) clouddevops.co.in

Keep blank to create a record for the root domain.

☐ Alias

Value [Info](#)

Record type [Info](#) **A - Routes traffic to an IPv4 address and some AWS resources**

Routing Policy: Select **Latency**.

Region: Choose the AWS region where the endpoint is located

Click **Create records**

EC2 Route 53

Create record [Info](#)

TTL (seconds) [Info](#)

Recommended values: 60 to 172800 (two days)

Routing policy [Info](#) **Latency**

Region [Info](#)

The Amazon EC2 region where the resource that you specified in this record resides. You can only create one latency record for each Amazon EC2 region. You aren't required to create latency records for all Amazon EC2 regions.

Health check ID - optional [Info](#)

Record ID [Info](#)

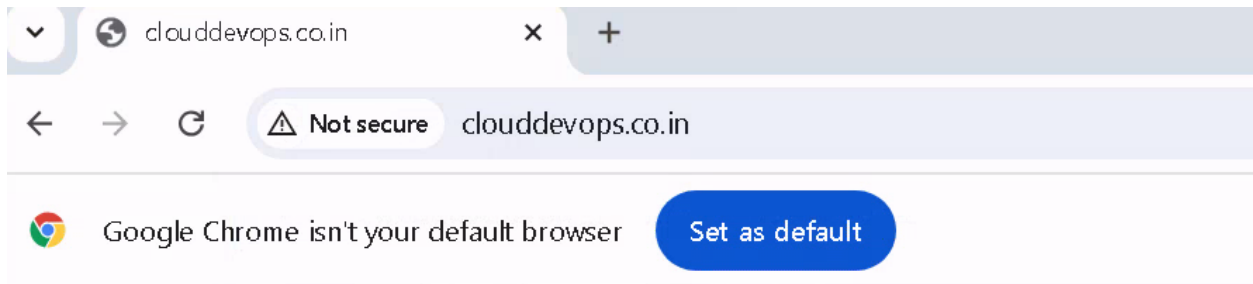
[Add another record](#)

[Cancel](#) [Create records](#)

Repeat the above steps to create a latency-based record for each additional region where you have resources.

Open a browser and go to your custom domain on the server from the **same location US WEST(Oregon)**. It should now route to your application via the load balancer.

From US WEST(Oregon) location server



EC2 instance from US(Oregon)